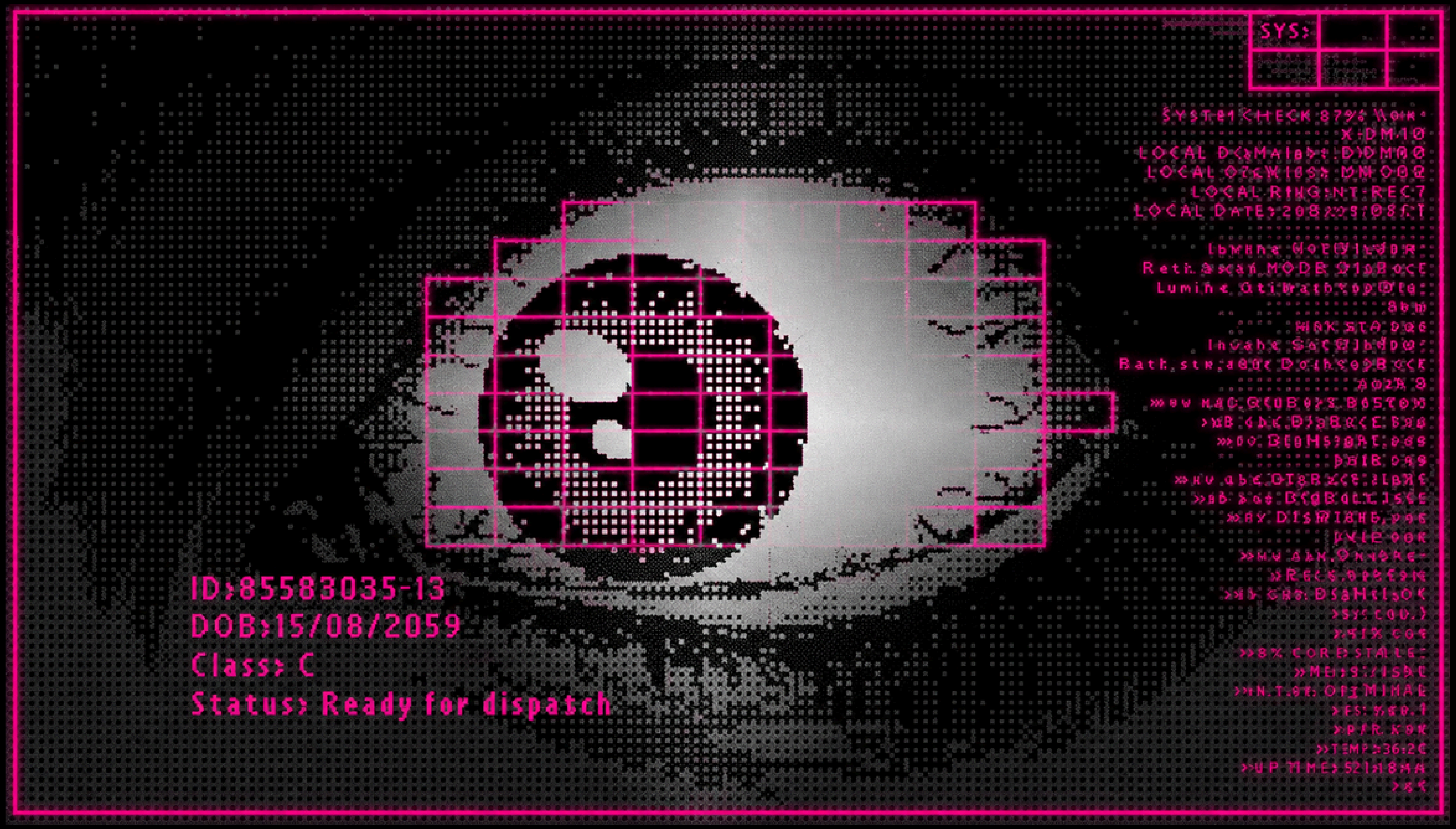
BEHIND THE IRIS: Biometric Recognition in Twin Subjects



Master’s Degree in Cybersecurity
Biometric Systems

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The Biometric Paradox: Epigenetics vs. Twins

- **The Epigenetic Premise**: Iris texture is rich, stable, and formed randomly during gestation. Classical systems assume this epigenetic randomness provides highly discriminative information, unaffected by DNA.
- **The Twin Problem**: Twins share identical genetic information and physical traits.
- Related Evidence:
  - *Hollingsworth et al*. showed human observers perceive visual similarity in genetically identical irises.
  - *Juniati et al*. demonstrated that increasing the proportion of twins in a dataset reduces overall recognition accuracy.
- **Our Focus**: *If humans see similarity, does a classical biometric pipeline detect it?*

Research Questions & Objectives

- **Impostor Behavior**: *Are twin comparisons (especially same-eye) inherently more similar than unrelated impostor comparisons?*
- **Identification Impact**: *Are twins the primary cause of errors in a closed-set identification scenario?*
- **Texture Validation**: *Does this "Twin Effect" persist across alternative texture descriptors?*
- **Failure Analysis**: *When the system fails, why does it fail? Is it due to genetic ambiguity, or limitations in template quality and segmentation?*

The CASIA-Iris-Twins Dataset

Initial audit

3183 raw images across 100 families.

- **Data Cleaning**: 2 exact duplicates removed to prevent artificially perfect genuine comparisons -> 3181 clean images.
- **Metadata Hierarchy**: Every image is strictly labeled by *family_id*, *twin_id, eye* (left/right), *subject_id*, and *iris_id*.
- **Biometric Trait**: The left and right irises of the same subject are treated as distinct biometric identities (***iris_id***).

```
📁 CASIA-Iris-Twins Dataset [100 Families]
│
├──── 📁 Family_01
│    │
│    ├──── 📁 Twin_1 (subject_id)
│    │    ├──── 📁 Left_Eye  ➔ (iris_id: 01_1L)
│    │    │    ├──── 📄 S3011L01.jpg
│    │    │    └──── 📄 S3011L02.jpg
│    │    │    └──── 📄 ...
│    │    │
│    │    └──── 📁 Right_Eye ➔ (iris_id: 01_1R)
│    │        ├──── 📄 S3011R01.jpg
│    │        └──── 📄 S3011R02.jpg
│    │        └──── 📄 ...
│    │
│    └──── 📁 Twin_2 (subject_id)
│        ├──── 📁 Left_Eye  ➔ (iris_id: 01_2L)
│        └──── 📁 Right_Eye ➔ (iris_id: 01_2R)
│
└──── 📁 Family_02 ...
```

Modular Pipeline Architecture

1 Quality Audit

Preliminary analysis of contrast, focus, and brightness for diagnostic purposes.

2 Daugman-Style Segmentation

Geometric estimation of circular pupil and iris boundaries and strict rejection of geometric anomalies.

3 Rubber Sheet Normalization

Transformation of the annular iris into a fixed-coordinate rectangular map (64x512).

4 Gabor Feature Extraction

Exclusive isolation of lateral regions to remove noise and generation of balanced binary templates.

5 Masked HD Matching

Comparison of shared valid bits only, with angular shift compensation for eye rotation.

System Engineering: Tools & Design Choices

Component	Standard Libraries	Our Custom Code
Data & Files	Pandas, Pathlib	Twin-specific dataset hierarchy
Image Math	OpenCV, NumPy	Daugman logic, Rubber Sheet mapping
Biometrics	-	Mask generation, Gabor filters, Masked HD

Our Engineering Philosophy:

- **Quality Over Quantity ➔** We intentionally sacrificed dataset size to guarantee absolute geometric accuracy.
- **Zero-Tolerance for Noise ➔** Custom masking strategies were built to completely isolate the iris from biological artifacts (eyelids/lashes).
- **Unbiased Metrics ➔** The mathematical matching was designed to compare only perfectly reliable regions, avoiding artificial penalties.
- **Targeted Evaluation ➔** Pair generation was highly controlled to ensure the twin analysis was computationally feasible without losing a single genetic comparison.

Image Quality Audit & Distributions

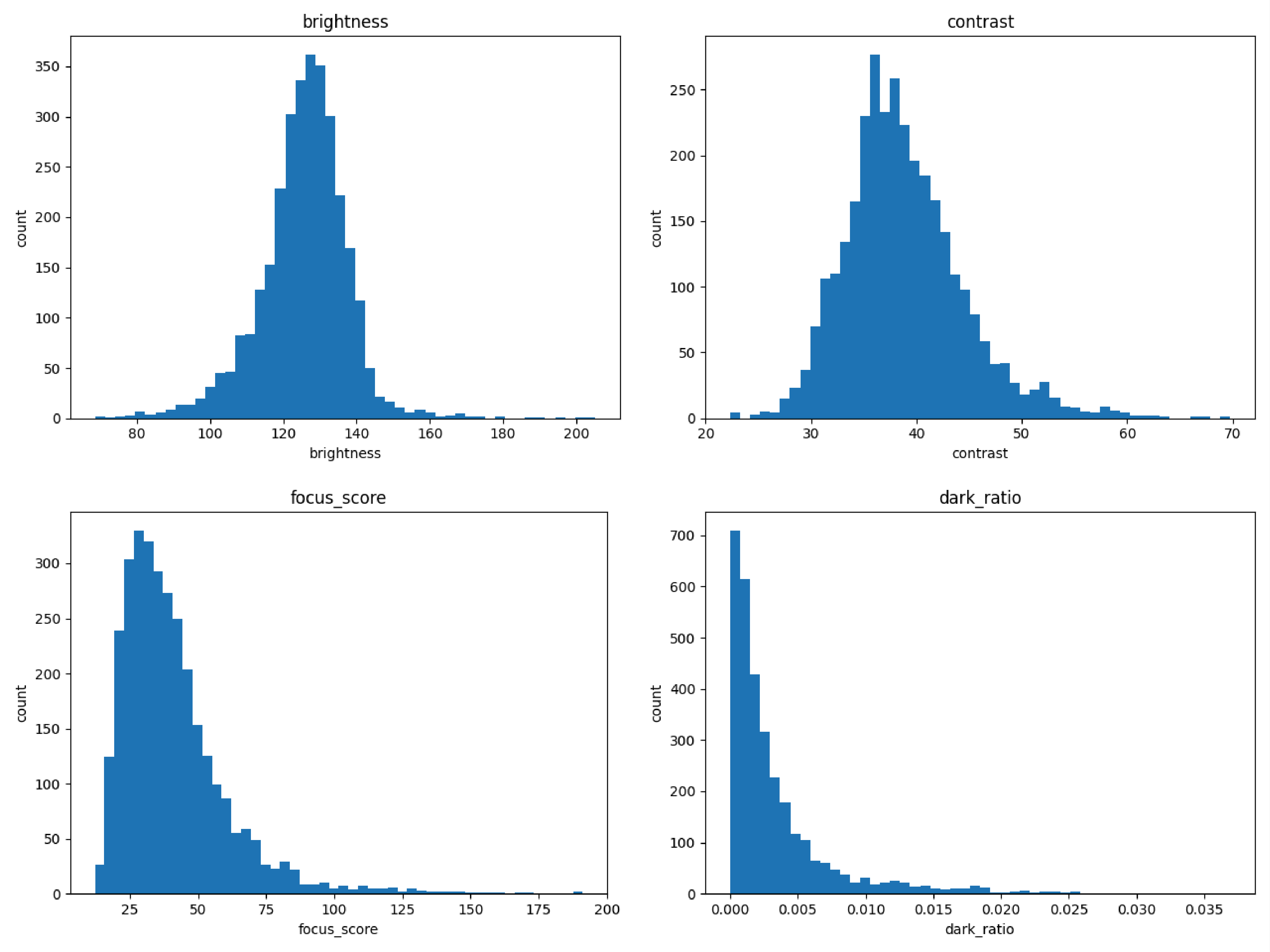
- **Goal**: Compute quality metrics (Brightness, Contrast, Focus, Extreme Pixels) for later failure analysis.

- Most images show stable brightness and contrast.

- **The Outliers**: Long tails in focus and dark/bright ratios indicate problematic samples that we intentionally kept.

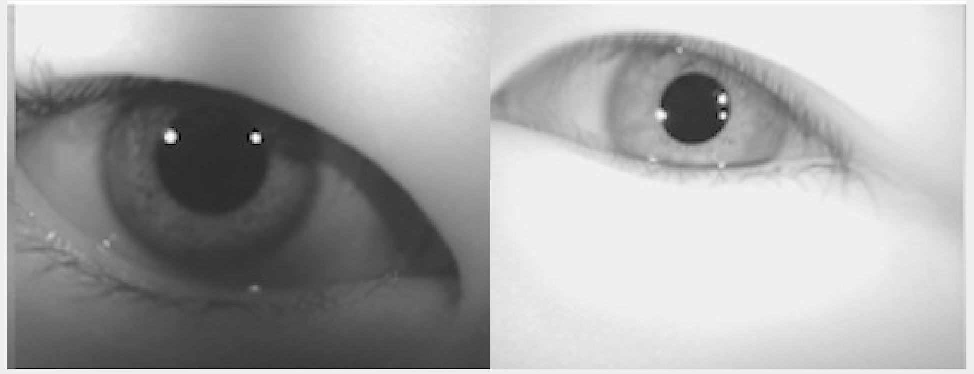
Why keep the outliers?

- **No arbitrary thresholds**: Avoided human bias in data filtering.
- **Algorithmic rejection**: Let the Strict Segmentation naturally discard unusable images.
- **Real-world testing**: Preserved hard cases for realistic evaluation and Failure Analysis.



The Impact of Quality Extremes

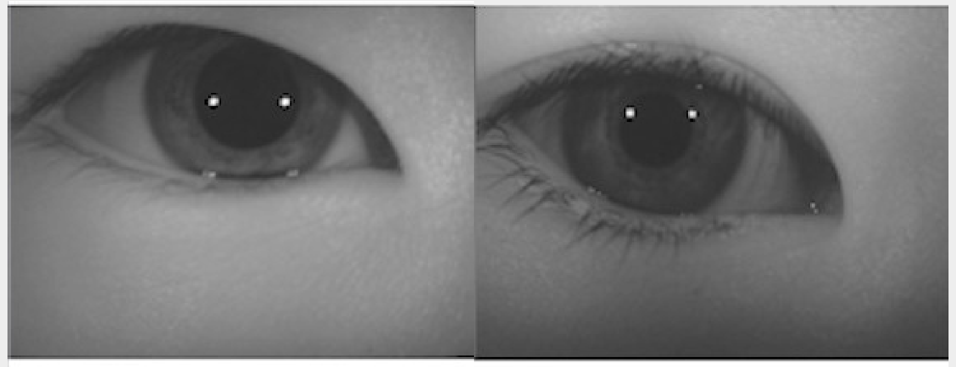
Bright ratio



High bright ratio:

- reflections/saturation hide iris texture
- mask removes more pixels

Dark ratio



Low brightness / high dark ratio:

- pupil and iris boundaries become harder to detect
- segmentation may fail or shift

Contrast:

Low contrast:

- iris texture is less visible
- Gabor/LBP features become weaker

Focus

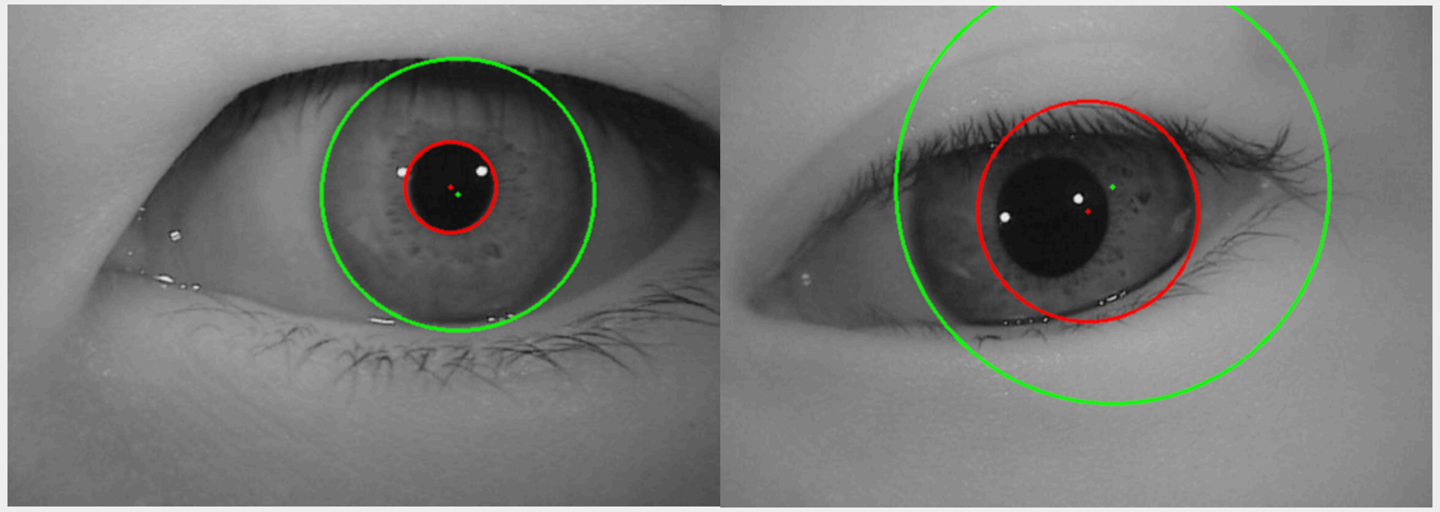
Low focus score:

- blurred texture
- same iris can produce higher Hamming Distance

These values were not used to discard images automatically, but to explain segmentation, matching and identification failures.

Iris Segmentation:

From Broad Acceptance to Strict Filtering



conf=high | ratio=3.00

conf=low | ratio=1.97

Method	Accepted	Rate
V1 diagnostic	744	23.29%
Custom V2	3110	97.77%
Daugman-style initial	2196	69.03%
Strict Daugman-style	**1465**	**46.05%**

- **Trade-off**: We evaluated multiple approaches to balance dataset size and geometric accuracy.
- **Final Choice**: We opted for the Strict Daugman-style, sacrificing over half the dataset (46.05% retained) to guarantee zero tolerance for misaligned boundaries.

Daugman-style Strict Segmentation

- The positions and sizes of the pupil and iris boundaries are estimated using a**utomatic segmentation algorithms**, specifically Daugman’s integro-differential operator for circular boundary detection and active contour (snake) models for refining the iris and pupil edges.

- **Strict quality checks** are applied:

    - **ratio** between pupil and iris radii, to verify realistic proportions

    - **distance** between pupil and iris centers, to exclude abnormal misalignments

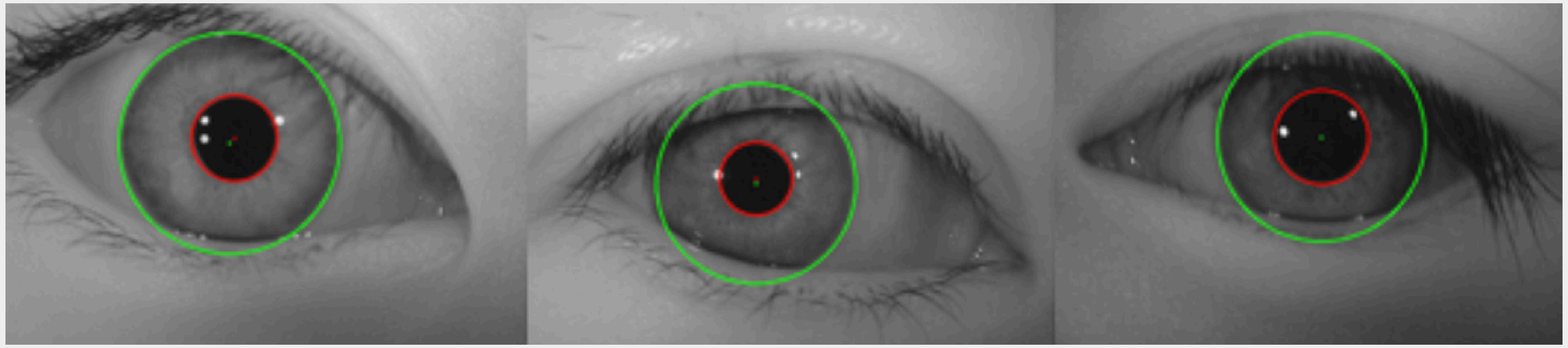
    - **iris score**, which measures the overall quality of the segmentation

    - **pupil darkness** level, to ensure proper contrast

    - **contour plausibility**, to eliminate irregular or incomplete segmentations

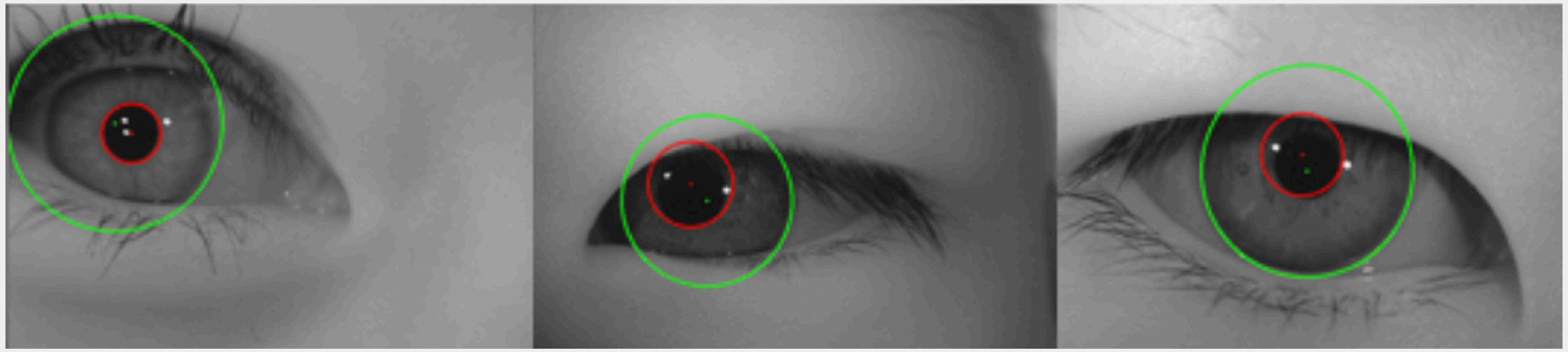
Segmentation examples

Accepted



- **Accurate Localization**: Precise pupil/limbus detection, robust to reflections.
- **Anatomic Limits**: Circular models inevitably capture eyelids and eyelashes.
- **Masking Required:** Circular boundaries include eyelids, making the next masking step crucial.

Rejected



- **Visual vs. Metric**: Visually "ok" images are rejected due to hidden geometric flaws.
- **Offset Errors**: Centers are severely misaligned (e.g., offset > 20px).
- **Ratio Errors**: Iris-to-pupil ratio exceeds the 3.1 limit, capturing sclera and skin.

Final strict subset

1465 images
99 families
313 iris identities

- Visually acceptable boundaries often hide geometric flaws
- Flawed boundaries cause the Rubber Sheet model to unwrap skin, eyelashes, and sclera instead of the iris.
- Processing non-iris texture artificially inflates genuine Hamming Distances, invalidating the twin analysis.
- We prioritized strict biometric integrity over dataset size, retaining only the most reliable **46.05%** of images.

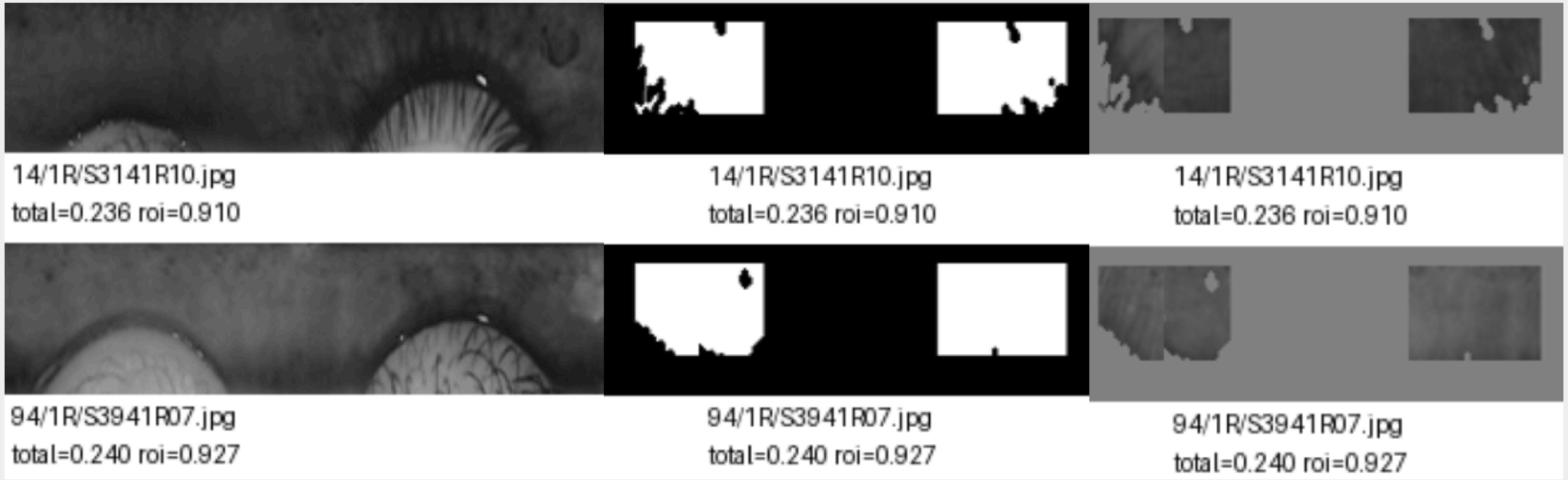
Normalization and Masking

- **Rubber Sheet Model**: Converts the annular iris region into a fixed-size rectangular strip (64 × 512 pixels).

- **Conservative v3 Mask**: Discards top/bottom areas and keeps only reliable lateral iris regions to avoid eyelids and eyelashes.

- **Mean Total Valid Ratio**: 0.2365 (we use **~23%** of the normalized strip).
  - *Design Choice: Prioritizing reliable iris regions over raw coverage to minimize noise.*

- **Mean ROI Valid Ratio**:
  0.9116 (ensuring high purity in the selected regions).



Gabor-style Binary Feature Extraction

- **Input**: normalized iris strips + masks
- **Four Gabor-style filters**:
  - 0° / 90°
  - wavelength 8 / 16
- **IrisCode Generation:** Binary code generated from filter response sign**.**
- **Mask** propagated to feature level.
- All **1465** images successfully converted.

Feature OK	1465
Feature Failed	0
Mean valid bit ratio	0.2151
Mean bit-one ratio	0.5079

Pair Generation

Relation Class	Possible Pairs	Selected Pairs
Genuine	3,984	**3,984**
Same Subject (Diff. Eye)	3,295	**3,295**
Twin (Same Eye)	3,102	**3,102**
Twin (Cross Eye)	3,008	**3,008**
Unrelated (Impostors)	1,058,991	**50,000**

- **Input**: 1465 feature templates.
- **Combinatorial Explosion**: 1,072,380 total possible unordered pairs.
- **Final Selected Protocol**: 63,389 total pairs evaluated.

Masked Hamming Distance Matching

- **Mask Intersection**: A bit is compared only if it is valid in both mask A and mask B.

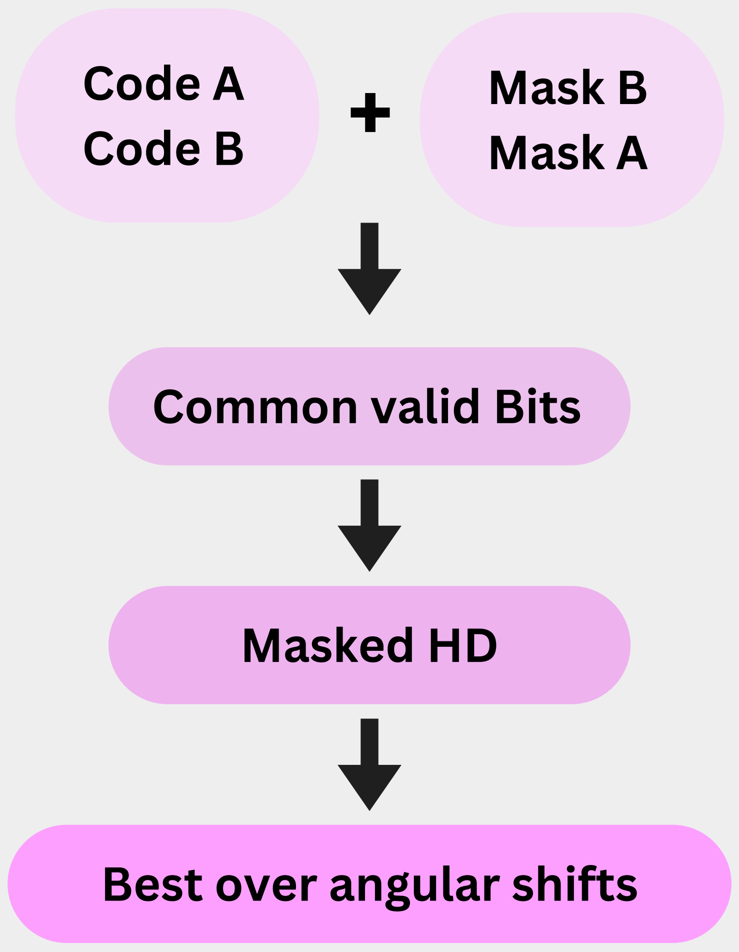
- **Distance Metric**: Measures the fraction of disagreeing bits *(Lower HD = higher similarity)*.

- **Angular Shift Compensation**: Accounts for eye/head rotation by testing multiple angular shifts:

$$s \in \{-16,-14,\ldots,14,16\}.$$

- **Final Score**: The minimum Hamming Distance obtained across all shifts.

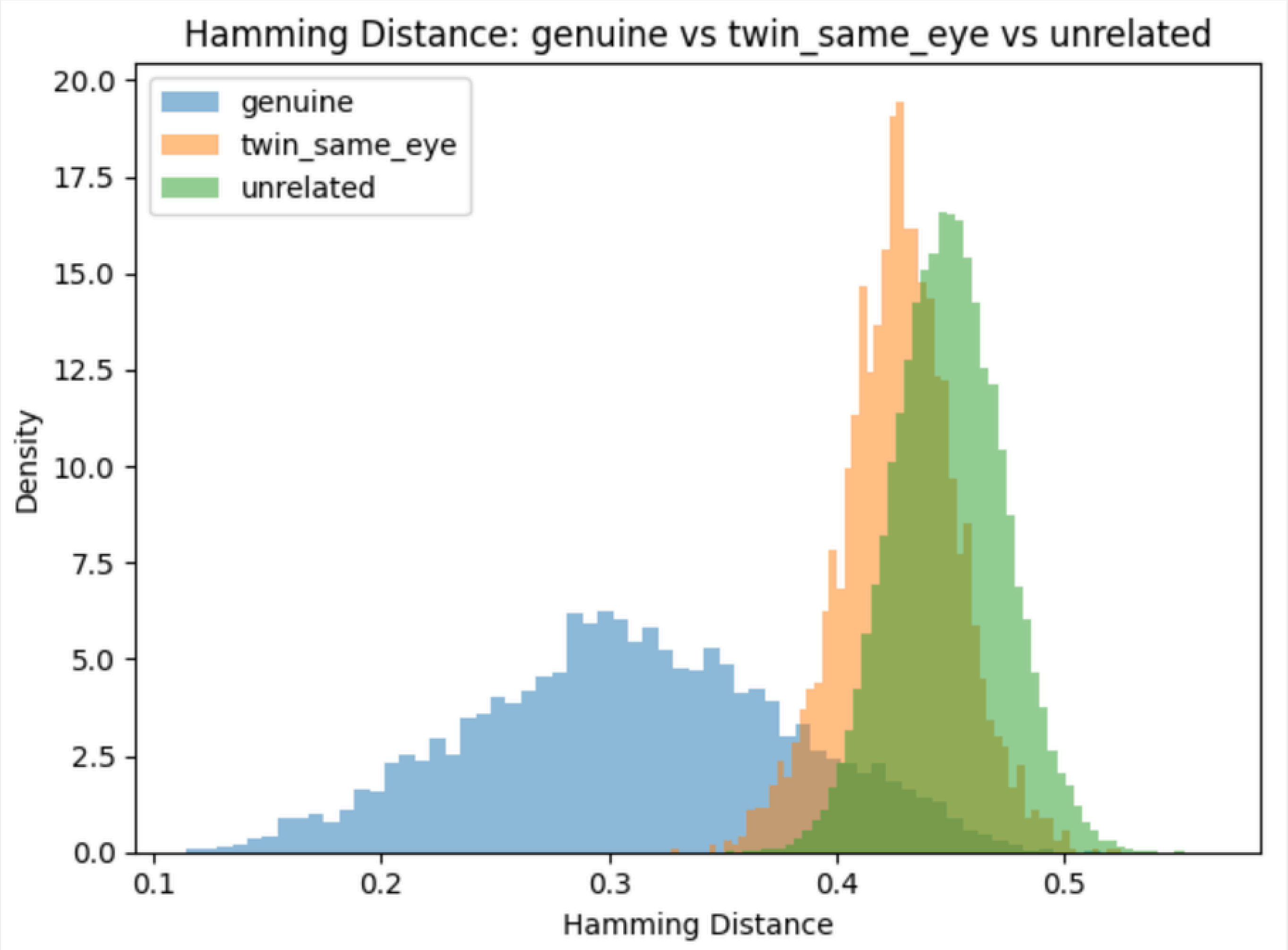
$$\textbf{HD} = \frac{\| (\text{codeA} \oplus \text{codeB}) \cap \text{maskA} \cap \text{maskB} \|}{\| \text{maskA} \cap \text{maskB} \|}$$



Verification Results: Score Distributions

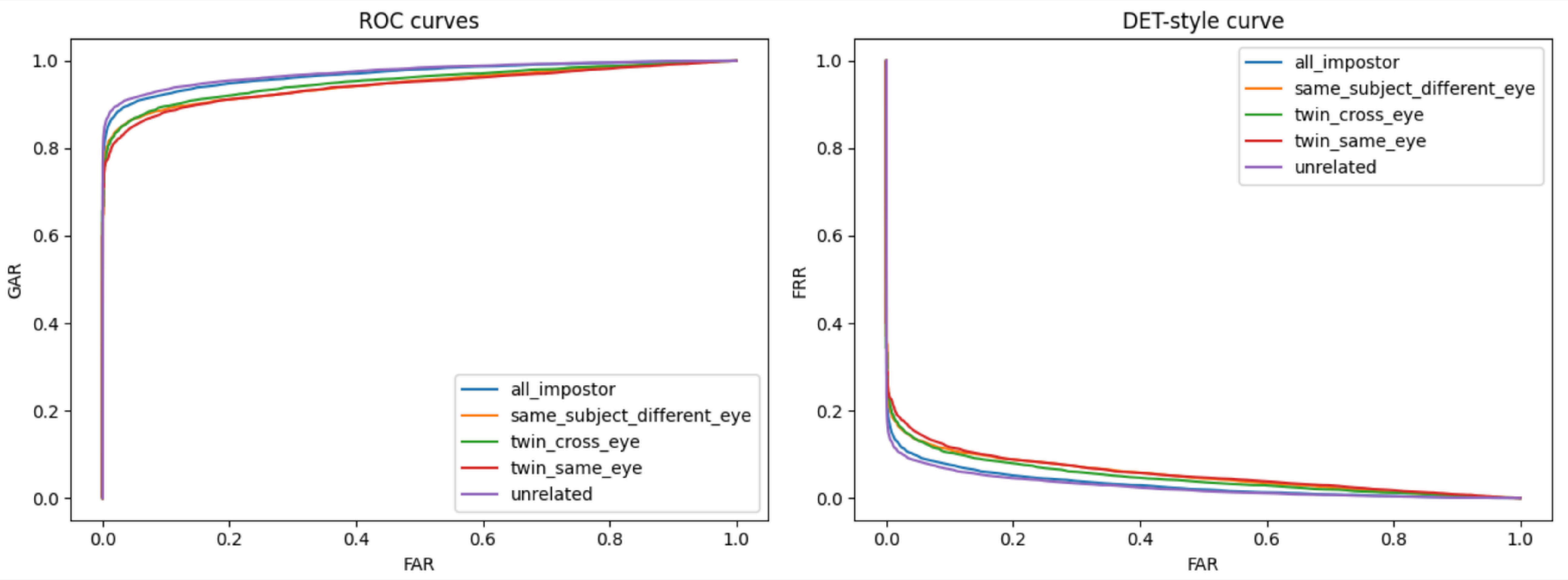
- **Clear Separation**: Genuine pairs are distinctly separated from all impostor classes.
- **The Twin Effect**: Twin pairs (especially same-eye) generate lower distances than unrelated pairs.

Relation	Mean HD
genuine	0.3098
twin_same_eye	0.4276
same_subject_different_eye	0.4301
twin_cross_eye	0.4331
unrelated	0.4492



Verification Performance: The Impact of Twins

- **Baseline Performance**: High accuracy against standard unrelated impostors (EER 7.55%, AUC 0.9735).
- **The Twin Penalty**: Verification becomes noticeably harder when the impostor is a same-eye twin (EER jumps to 11.39%, AUC drops to 0.9425).
- The ROC curve visually confirms that twin impostors shift the error trade-off, verifying the "**Twin Effect**" quantitatively.

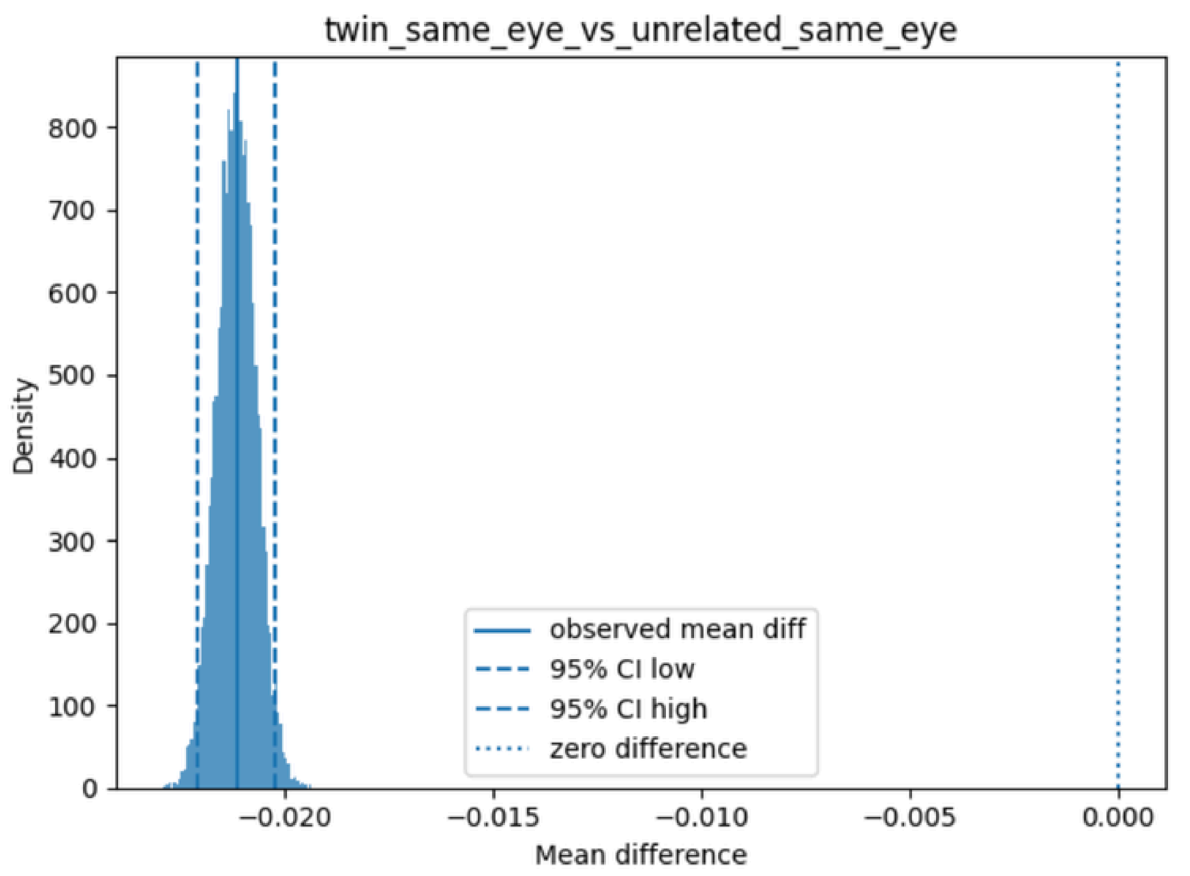


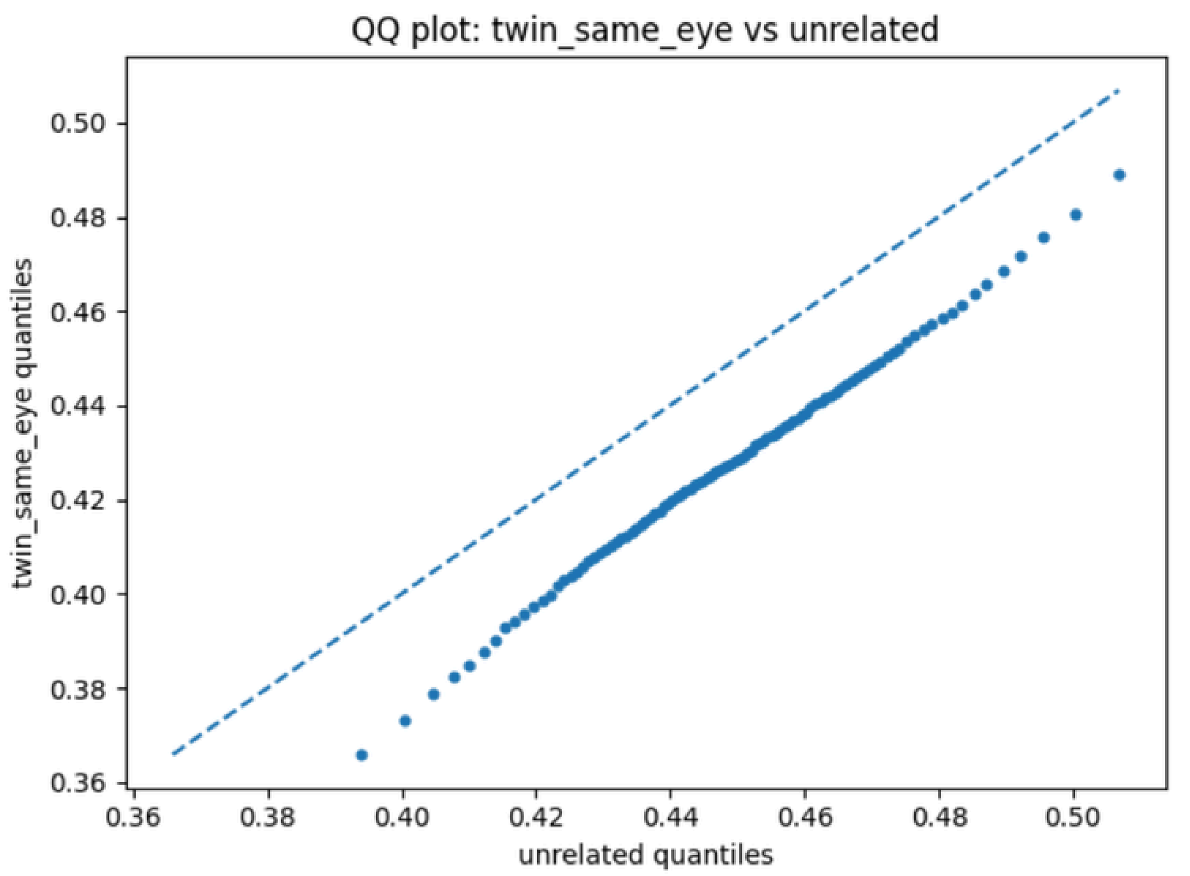
Impostor Set	EER	AUC
Unrelated	**0.0755**	**0.9735**
All Impostor	0.0817	0.9693
Twin (Cross Eye)	0.1041	0.9518
Same Subject (Diff. Eye)	0.1067	0.9459
Twin (Same Eye)	**0.1139**	**0.9425**

Statistical Evidence of Twin Similarity

- Our goal is to prove that the visual shift in twin scores is statistically robust.
- **Effect Size (Cohen's d)**: Measures the magnitude of the difference between distributions.
- ***Key Finding 1***: Twin pairs are consistently closer than unrelated pairs (Cohen's d = -0.8722 for same-eye).
- ***Key Finding 2***: The twin effect is measurable but much smaller than the genuine-vs-unrelated separation (Cohen's d = -4.6587).

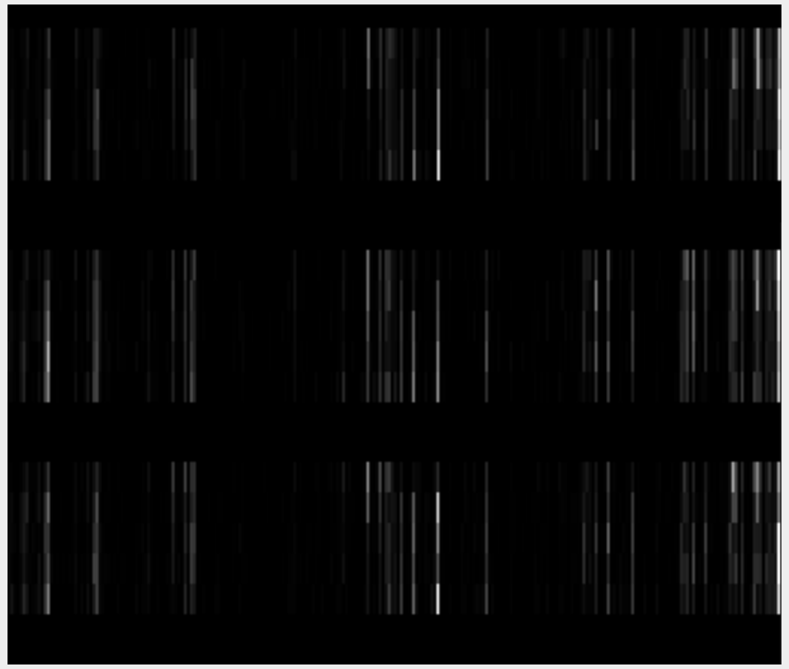
Comparison	Mean Diff.	Cohen's d
Genuine vs Unrelated	-0.1395	-4.6587
Twin All vs Unrelated	-0.0189	-0.777
Twin Same-Eye vs Unrelated Same-Eye	**-0.0212**	**-0.8722**
Twin Cross-Eye vs Unrelated Cross-Eye	-0.0165	-0.6797
Twin Same-Eye vs Twin Cross-Eye	-0.0055	-0.2218





Cross-Validation: LBP and Blob/LoG Descriptors

Metric	Genuine	Twin (Same Eye)	Twin (Cross Eye)	Unrelated
LBP (χ2)	0.4249	**0.5325**	0.5458	0.6395
Blob (Euclidean)	0.6603	**0.9296**	0.9274	1.0727
Blob (Cosine)	0.4538	**0.7797**	0.8	1.0025



lbp previews

- The goal is to verify if the "Twin Effect" is merely an artifact of the Gabor-style matcher.
- *Alternative Descriptors*: Evaluated Local Binary Patterns (**LBP**) and **Blob/LoG** multi-scale features.
- Consistent Trend: Across all metrics, twin distances constantly sit between genuine and unrelated baselines.
- The twin similarity is an intrinsic physical property of the iris texture, independent of the algorithm used.

Closed-set identification

In closed-set identification, each probe is compared against **all** gallery templates.

What is it?

The correct identity is expected to be in the gallery.

How it works?

Probe iris → Compare with gallery → **Ranked list** → Correct identity rank

Metrics

The system ranks all candidates by lowest **Masked Hamming Distance**.

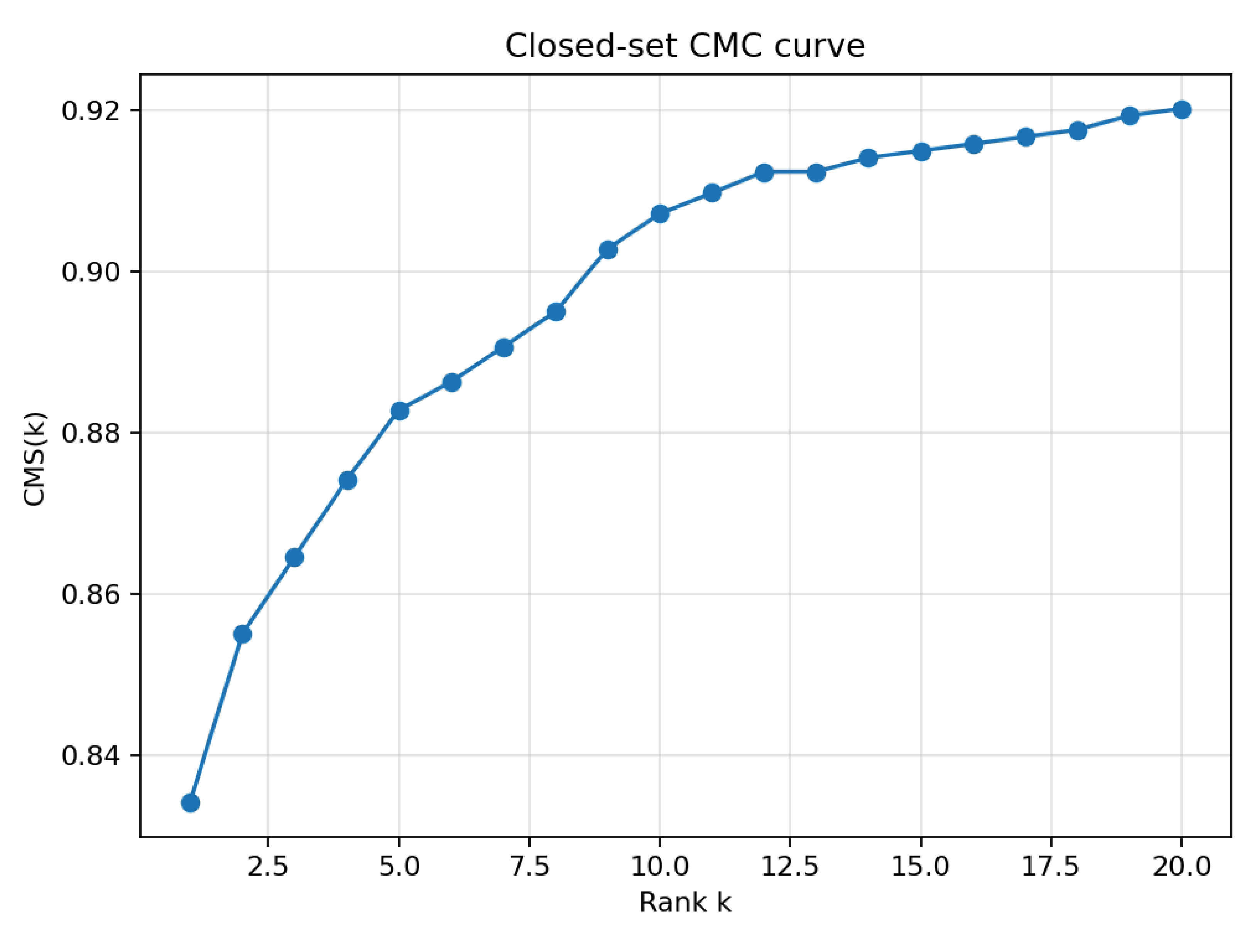
Evaluated via Rank-1 Accuracy and Cumulative Match Curve (CMC).

Dataset and gallery construction

Element	Value
Input feature images	**1465**
Gallery templates	**313**
Probe templates	**1152**
Gallery selection strategy	**Best valid bit ratio**
Probe-gallery comparisons	**360,576**
Failed comparisons	**0**

- **Quality-driven Enrollment**: For each of the 313 identities, we enrolled only the single template with the highest valid-bit ratio.
- **Real-world Simulation**: This mirrors operational biometric systems, which reject poor-quality samples during initial enrollment to ensure a reliable reference database.

How Fast Does the System Find the Right Iris?

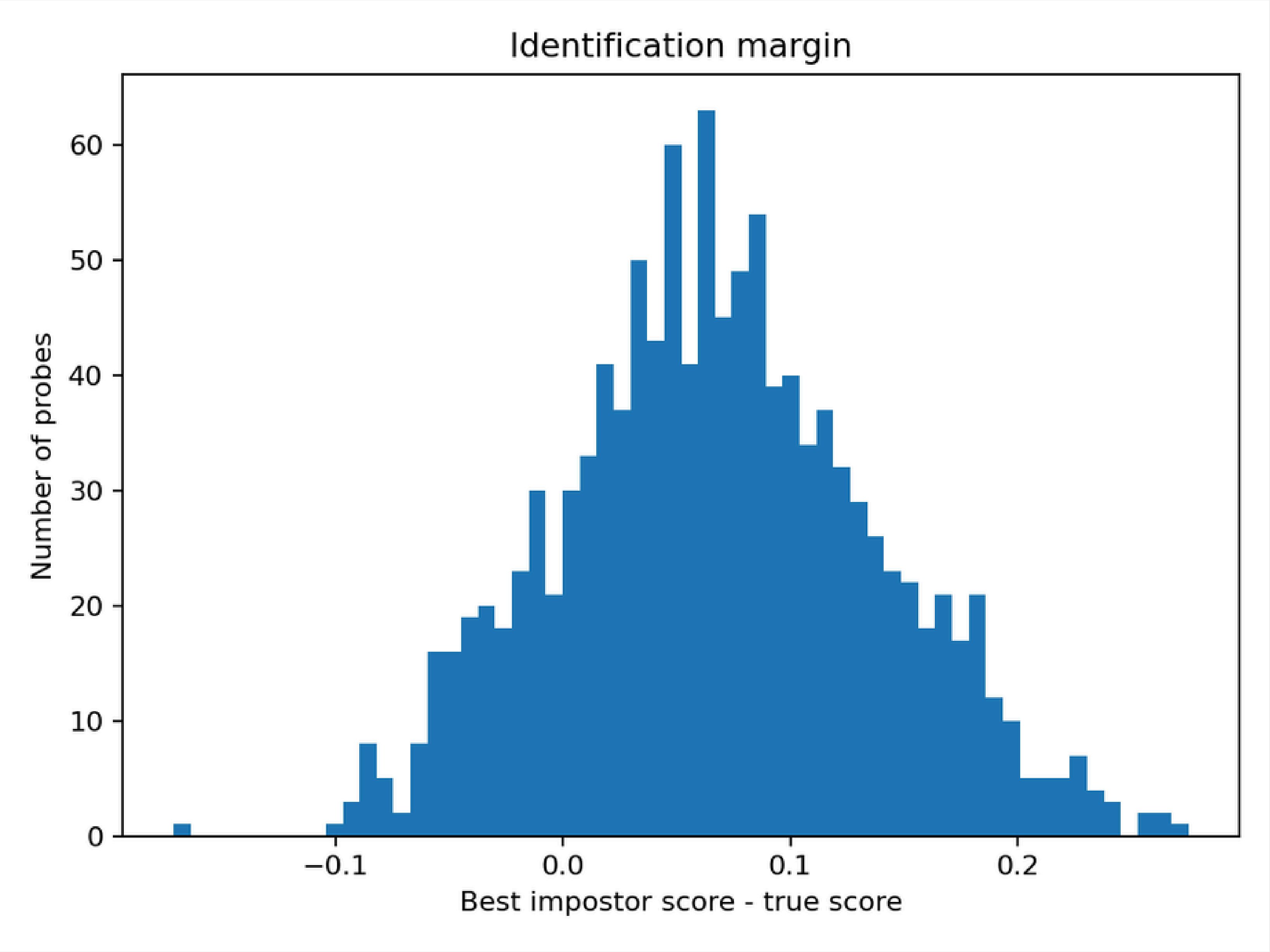


The curve shows the probability that the correct identity appears within the first k candidates.

Metric	Value
Rank-1 correct	952 / 1152
Rank-1 errors	200
CMS@1 / Rank-1	**0.8264**
CMS@2	0.849
CMS@5	0.8767
CMS@10	0.8993
CMS@20	0.9158

The system achieves strong closed-set identification performance, ranking the correct iris first in **83.42%** of attempts

Identification Margin



- Metric Definition
  - Calculated as:
    Margin *= Best Impostor Score - True Identity Score*
  - Quantifies decision reliability by measuring the gap between the true match and the closest competitor.
- **Positive Margin (> 0)**
  - Rank-1 Correct: The true identity is cleanly separated fr the closest impostor.
- **Negative Margin (< 0)**
  - Rank-1 Error: An impostor successfully bypassed the tru identity.

Twin-specific Identification Analysis

The Hypothesis

Verification results showed that twin irises produce slightly lower Hamming Distances than unrelated subjects.

This raises the key question:

Will genetic similarity break the identification pipeline?
Are twins the primary drivers of Rank-1 errors?

The Reality

Metric	Value
Twin present in gallery	93.32%
Best impostor is twin	4.86%
Rank-1 errors caused by twin	**9 / 200**
Twin in Top-5	13.72%
Twin in Top-10	21.27%
Median best twin rank	46

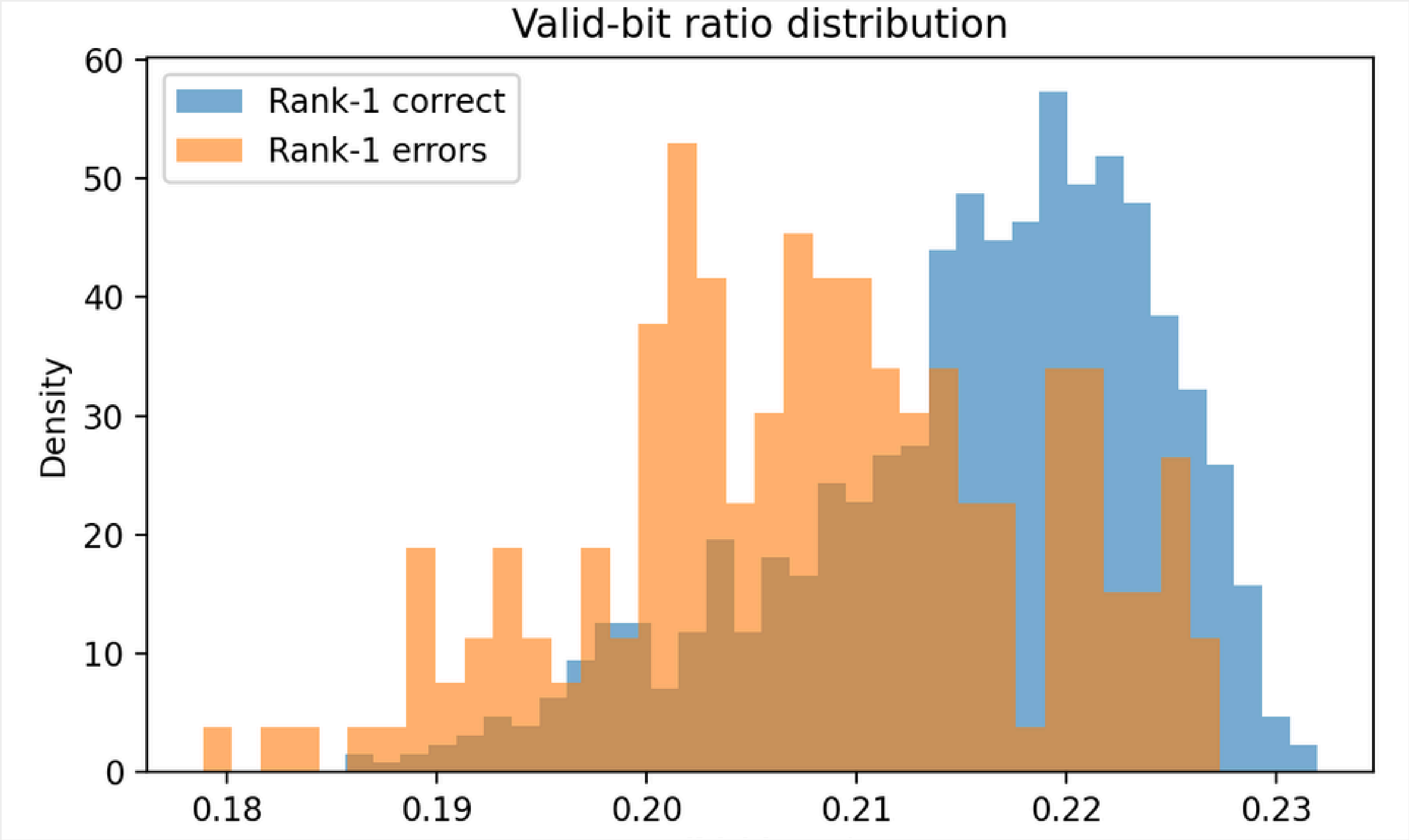
Verdict:
Twins increase local ambiguity, but they **are NOT** the dominant cause of Rank-1 errors.

Considering all the cases where the twin exists in the gallery, the closest twin is on median **rank 46.**

Quality-based Failure Analysis

Metric	Rank-1 correct mean	Rank-1 error mean
valid_bit_ratio	0.2156	0.208
focus_score	43.41	41.45
brightness	127.37	127.31
dark_ratio	0.0029	0.0029
bright_ratio	0.0013	0.0005

Errors are more associated with reduced usable iris texture than with simple illumination problems.

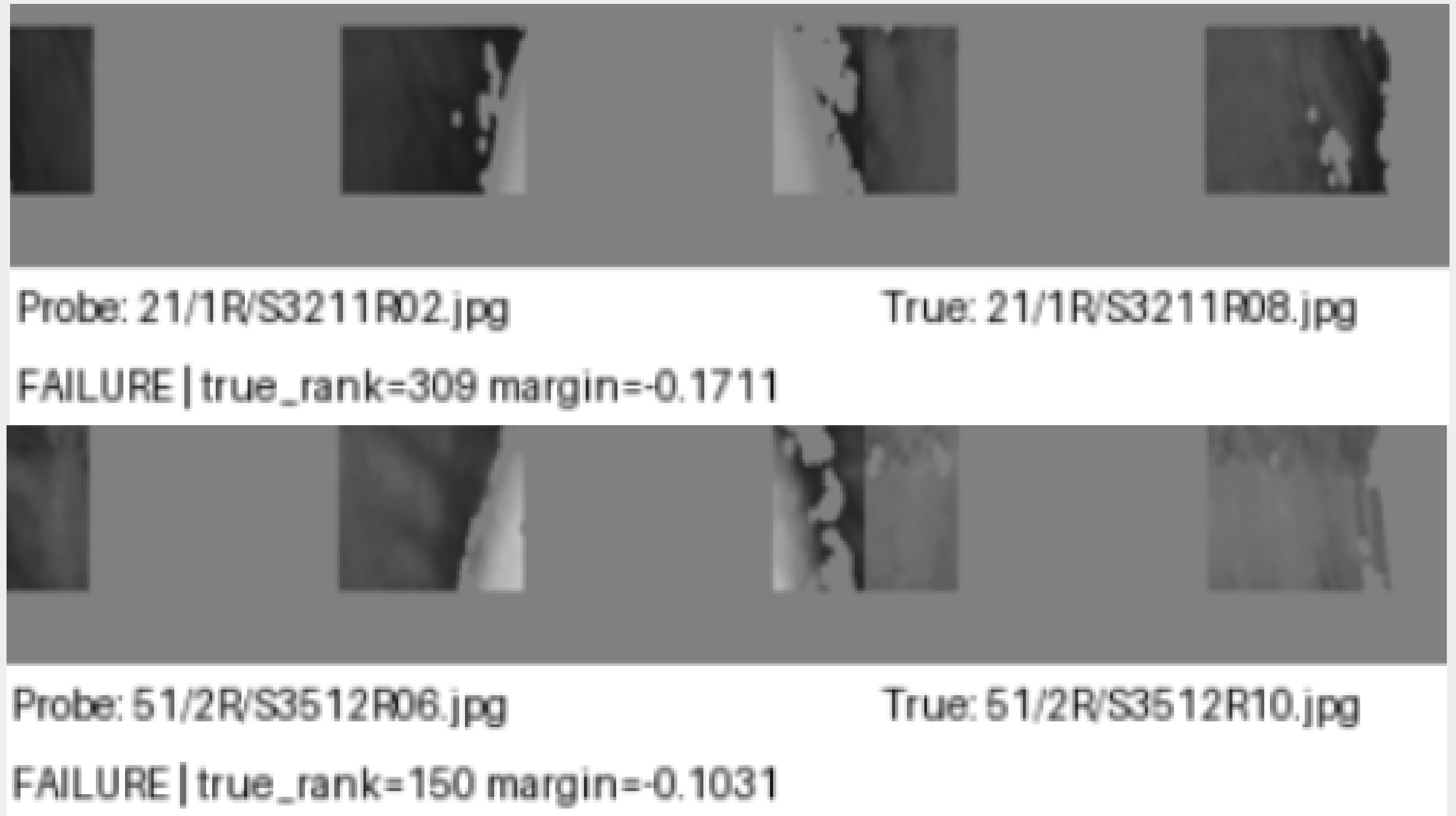


Rank-1 errors tend to have a lower valid-bit ratio than correctly identified probes.

How Template Quality Affects Identification Performance

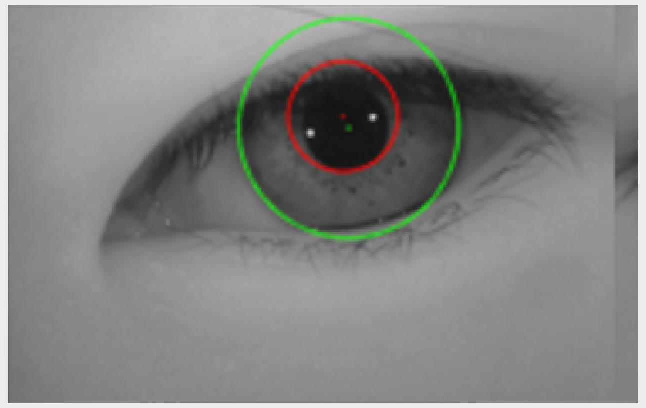
Successes

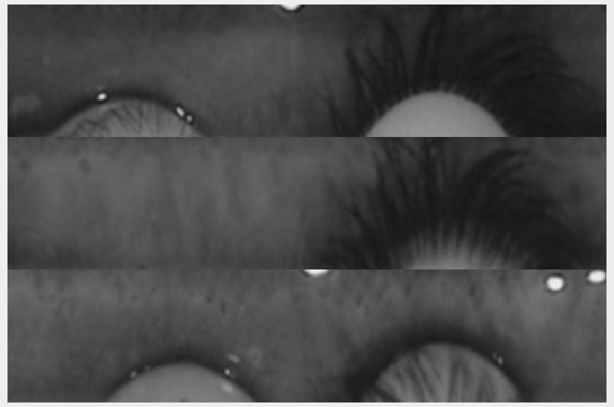
Failures



Limitations

Preprocessing Bottlenecks





- **Segmentation**: The Daugman-style procedure is a simplified implementation. Imperfect iris boundaries, eyelid occlusions, or reflections may persist.
- **Conservative Masking**: The v3 mask prioritizes reliability over raw coverage, significantly reducing the available iris texture to ~23.7% of the strip.

System Scope

- **Algorithmic Approach**: The system utilizes handcrafted Gabor-style features rather than commercial state-of-the-art recognition algorithms.
- **Evaluation Limits**: The current protocol covers verification and closed-set identification but does not implement open-set identification scenarios.

Statistical Validity

- **Comparison Dependency**: Pairwise biometric comparisons are not fully independent (the same image appears in multiple pairs), which may render statistical p-values overly optimistic.
- **Dataset Specificity**: Results are dependent on the labels, image quality distribution, and sample balance present in the CASIA-Iris-Twins subset.

Future Work

1 Robust Preprocessing

- **Goal**: Replace custom segmentation with advanced models.
- **Focus**: Better handling of eyelids, eyelashes, and off-axis acquisitions.
- **Result**: Increase valid bit ratio and template reliability.

2 Advanced Protocols & Zygosity

- **Zygosity Classification**: Distinguish between monozygotic, dizygotic, and unrelated subjects.
- **Multimodal Approach**: Fuse iris texture with periocular, face, or voice traits.
- **Explainable AI**: Use attention maps to interpret classification results.

3 Deep Learning Pipeline

- **Siamese Networks**: Replace handcrafted Gabor codes with learned embeddings.
- **Strategy**: Use triplet/contrastive loss for relation-aware classification.
- **Embeddings**: Push genuine pairs together and impostor pairs apart.

THANK YOU

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Repository:
https://github.com/samsungalaxys3neo/Casia_Iris_Twins.git